DATHEON INTERNSHIP

TASK 1
Exploratory Data Analysis Report

Superstore Sales Dataset

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Dataset	superstore_sales.csv
Tools	Python · Pandas · Matplotlib · Seaborn · Google Colab
Report date	12 November 2026

A concise analysis of sales performance, customer segments, regional trends, product profitability and discount patterns.

1. Project Objective

The objective of this project was to explore the Superstore sales dataset, identify patterns in customer purchasing behaviour, product and regional performance, sales trends and profitability, and translate the findings into practical business recommendations.

2. Data Preparation

The dataset was loaded and inspected using Pandas. Preparation included reviewing the dataset structure and data types, checking missing values and duplicate records, preparing date fields, and creating time-related fields for trend analysis.

3. Key Findings

- **Customer analysis:** The Consumer segment generated the highest sales, followed by Corporate and Home Office. Sean Miller was the top customer by sales.
- **Product analysis:** Technology led the three product categories in sales. Phones was the highest-selling sub-category. Canon imageCLASS 2200 Advanced Copier was the top product in the top-products analysis.
- **Regional analysis:** West recorded the highest regional sales and profit, while South recorded the lowest sales. New York City had the highest city-wise sales and Abilene the lowest.
- **Time analysis:** June recorded the highest monthly sales and February the lowest in the monthly summary.
- **Profitability:** Technology generated the highest category-level profit, while Furniture generated the lowest. Supplies, Bookcases and Tables were among the sub-categories with negative profit. Cubify CubeX 3D Printer Double Head was the largest loss-making product, with a loss of approximately $8,879.97.
- **Discount analysis:** The 10% discount level had the highest average profit. The 50% discount level had the lowest average profit, which was negative.

4. Selected Visual Analysis

The following charts support the main findings from the analysis.

Sales by Product Category

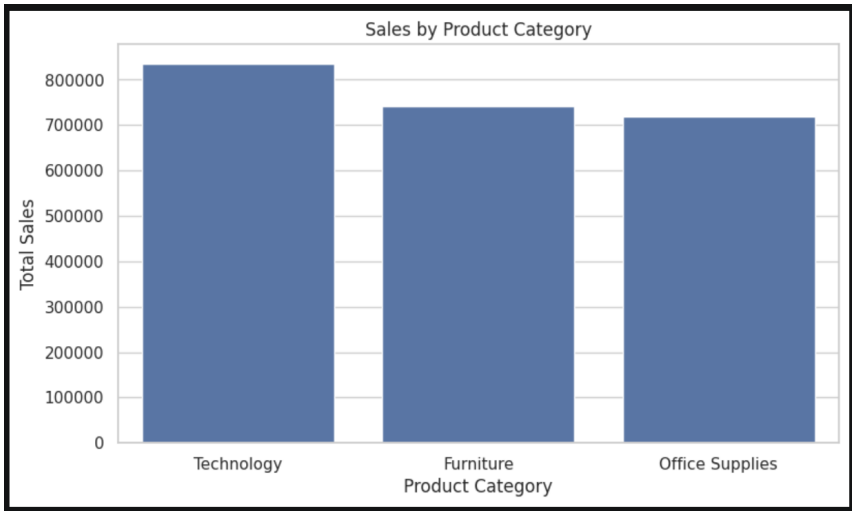


Figure 1. Technology recorded the highest category sales, followed by Furniture and Office Supplies.

Profit by Product Category

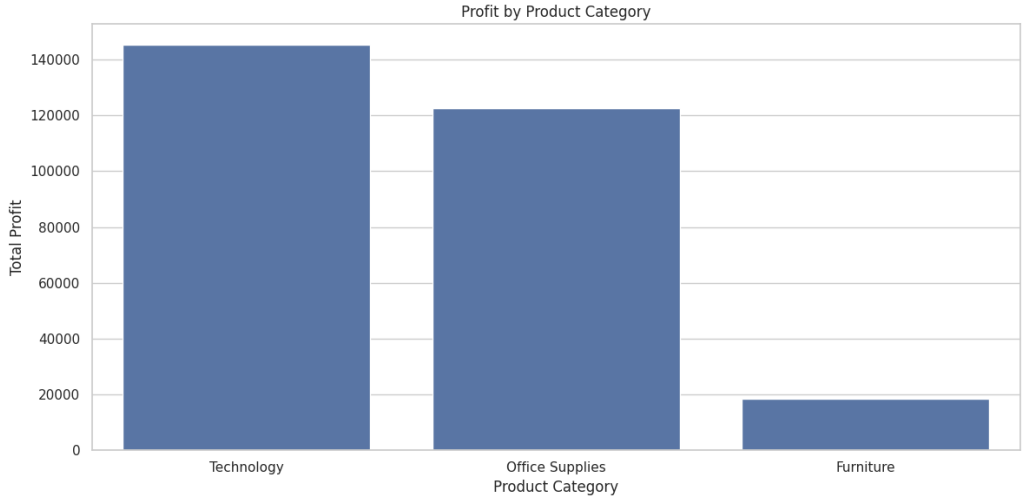


Figure 2. Technology generated the highest category profit; Furniture generated the lowest.

Regional Sales

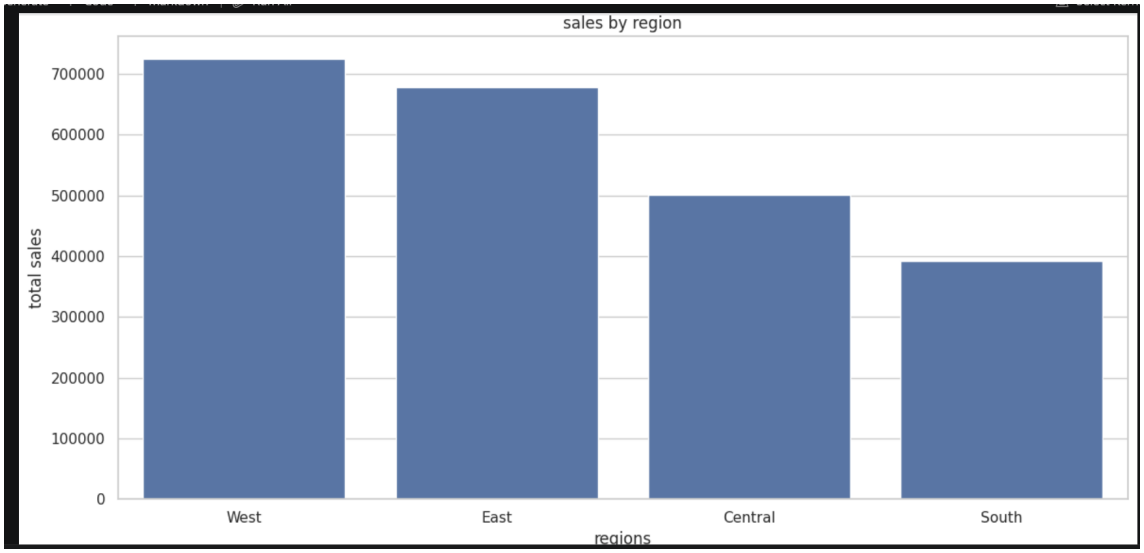


Figure 3. The West region led in sales, while the South region recorded the lowest sales.

Discount vs Average Profit

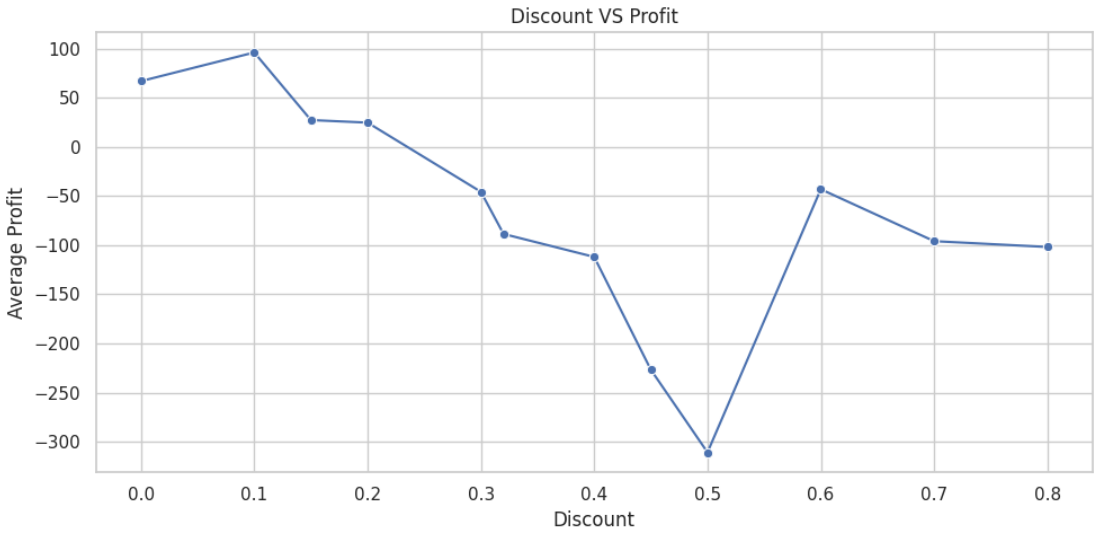


Figure 4. Average profit was highest at a 10% discount and lowest at a 50% discount. The relationship is associative and does not by itself establish causation.

Monthly Sales Trend

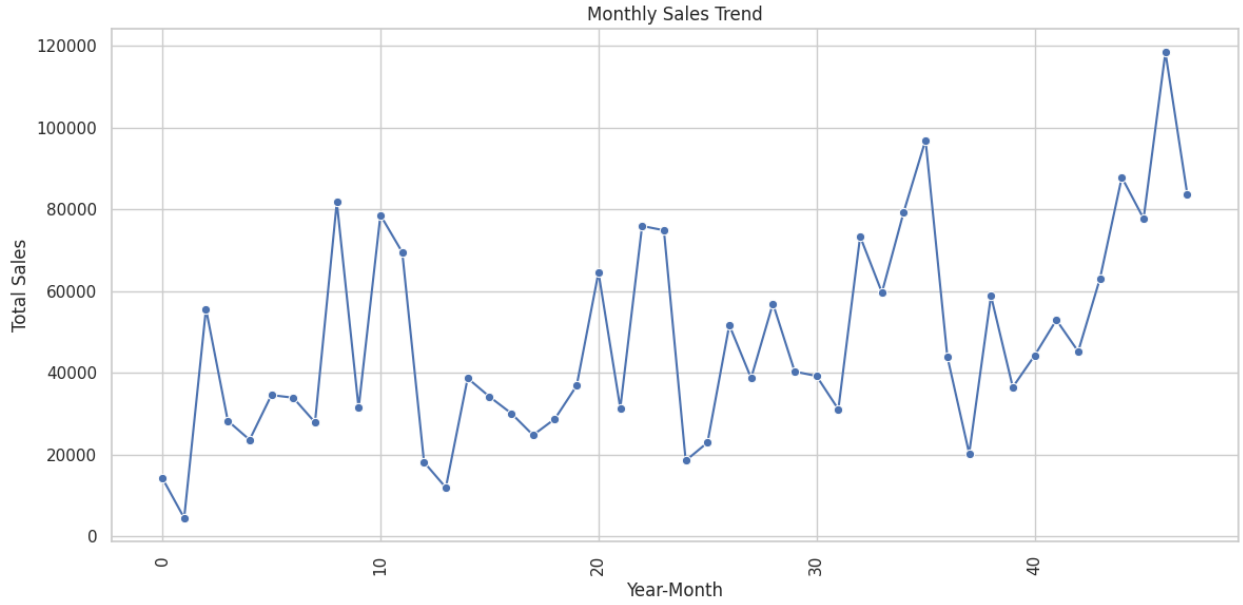


Figure 5. The chart displays monthly sales across the observed periods. The separate monthly summary identified June as the highest-sales month and February as the lowest.

5. Business Recommendations

- **Strengthen customer retention:** Continue engaging the Consumer segment and retain high-value customers.
- **Build on product strengths:** Monitor demand for high-performing Technology products while reviewing weaker products.
- **Improve regional performance:** Study practices in the West region and investigate opportunities to improve sales in the South.
- **Plan around sales patterns:** Use monthly trends to support inventory, staffing and promotional planning.
- **Review loss-making items:** Examine product pricing, costs, demand and discounting for products and sub-categories with negative profit.
- **Optimize discounts:** Reassess high discount levels, particularly 50%, to reduce the risk of unprofitable transactions.

6. Conclusion

The exploratory analysis highlighted strong performance in the Technology category and West region, identified seasonal variation in sales, and revealed loss-making products and sub-categories. The discount analysis also showed that the 50% discount level had negative average profit. These insights can help guide product reviews, regional planning and more disciplined discount strategies. The analysis demonstrates how Python-based EDA can turn raw sales data into useful business insights.

Prepared as part of Datheon Internship – Task 1.